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Streamlining cross-border shipping of live invertebrates

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Summary

The logistics of shipping live invertebrates should be straightforward: it requires timely delivery and packaging that ensures survival and confinement. The packaging is the responsibility of the shipper, whose interest is to maintain the product's quality during shipment and to ensure that no specimens escape. Timely delivery relies on the ability of the shipping agent to organise an efficient route of transport and carriers to fulfil their function effectively.

This article explores this underserviced sector. While similar logistics capacity exists for other goods that require fast delivery (such as vaccines and fresh food), stakeholders who require live invertebrate shipping often have difficulties in finding transporters able or willing to handle such services. The authors bring examples from biocontrol, showing the current complexity and inconsistency of logistics. For some countries and stakeholders, this issue can be a significant barrier to the growth of a sustainable biocontrol sector.

The authors also explore misconceptions (about packaging, liability and paperwork) and unclear rules (such as requiring generic veterinary certificates that are rarely relevant for invertebrates) that may cause express courier companies to refuse carrying live invertebrates. These issues often result in packages not being handled as a priority during transport connections or customs clearance, and significant delivery delays.

The authors propose improvements that could streamline transport through changes that fit within existing shipping processes. This article is furthermore intended as a call to transport and inspection stakeholders to use the existing guidance and other resources to support this underdeveloped sector more effectively.

Keywords

Biocontrol – Brokers – Express carriers – Live invertebrate – Logistics – Shipment.

Live invertebrate shipping: an underserviced market sector

Invertebrates are shipped across the world for various uses, from research activities to commercial agriculture, environmental protection and human- and animal-health applications. Shipments can be sporadic and consist of a single specimen or may be frequent and consist of large consignments of millions of specimens (Table I).

In shipping invertebrates, the biocontrol industry currently faces major challenges that impair continued proliferation of biocontrol and research activities. Challenges include timorous transport stakeholders, inefficient routes, inefficient handling and ambiguous and contradictory regulations between countries. The results can be prohibitive costs, difficulty providing authorities with the requested documentation, delayed delivery and damage or loss of the insects. Failure to ship can have enormous impacts on field control activities that correspond with pest outbreaks. In the long term, unreliable shipping can inhibit the expansion of biocontrol-based crop protection and result in continued reliance on conventional pesticide.

Over 1,000 biological control products are supplied all over Europe. Approximately 140 await registration, and a

Table I

Categories of live invertebrate biological control agents and other invertebrates shipped

Categories of invertebrates, mites and nematodes		Field of use	Stakeholders	Typical volumes	Typical frequency	Risk of shipment failure
IBCAs	Classical biocontrol	Agriculture Environment protection	Commercial Sustainable agriculture Environmental health	Medium (thousands of insects per consignment)	Weekly to monthly	Ineffective, inefficient crop protection service Use of pesticide
	Augmented biocontrol	Agriculture Environment protection	Commercial Sustainable agriculture Environmental health	High (millions to billions of insects per consignment)	Daily to weekly	Ineffective, inefficient crop protection service Use of pesticide
Animal health products (vector control)		Agriculture	Commercial Sustainable agriculture	Medium (thousands to millions of insects per consignment)	Daily to weekly	Ineffective, inefficient animal health protection service Use of pesticide
Sterile insects		Agriculture Environment protection Vector control programmes (mosquito, tsetse fly, etc.)	Commercial Sustainable agriculture Environmental health Public health	High (millions to billions of insects per consignment)	Weekly to biweekly	Ineffective, inefficient pest control Use of pesticide
Invertebrates for R&D activities		Agriculture Environment protection Human/animal health Rearing for food and feed	Research on pest management Research on transmitted pathogens/risks Colony starters	Low to medium (hundreds to millions of insects per consignment)	Irregular, a few times per year	Delayed, inefficient research Loss of unique samples Delay in colony establishment

IBCAs: invertebrate biological control agents
R&D: research and development

further 120 are in development [1]. The biocontrol industry is working hard to meet current demand and anticipates increased demand as a result of the European Green Deal, which aims to reduce pesticide use by 50% by 2030. According to Lamichhane *et al.* [2], the inaccessibility of biological solutions is already a significant constraint to growers who wish to find substitutes for conventional pesticides.

With growers of all sizes distributed across Europe and globally, efficient transport is crucial for the timely delivery of live invertebrate biological control agents (IBCAs), also known as macrobial products. Large biocontrol industry players rely on well-developed distribution channels and partners. Smaller suppliers, start-up businesses, and research and development (R&D) teams that require small or infrequent shipments often do not have sufficient economies of scale compared to their larger counterparts. Complicating matters, shippers making infrequent shipments sometimes lack skills and knowhow regarding packaging, documentation and other requirements. Out of precaution, local express carriers will sometimes refuse services when they lack protocols for the safety of such consignments.

The International Standard for Phytosanitary Measures (ISPM) No. 3 [3] guidelines have been developed to ensure seamless and safe export, shipment, import and release of IBCAs. They are widely recognised by national plant protection organisations (NPPOs). Additionally, the European and Mediterranean region follows the European and Mediterranean Plant Protection Organization (EPPO) PM 6/3 guidelines [4]. However, a disconnect exists between these guidelines and express carriers' understanding of safety and their subsequent willingness to transport IBCAs. Key deterrents include express carriers' mistaken beliefs that invertebrate transport should follow complex live animal transport regulations.

The COVID-19 pandemic has exacerbated problems arising from the dependence of biocontrol programmes and research activities on efficient cross-border logistics.

This paper is aimed to bring the attention of logistics stakeholders to the crucial part they can play in both the sustainable production of quality food and the protection of the environment by reducing pesticide externalities. A new partnership between shipping companies and live insect trade clients of all sizes and capacities could significantly contribute to meeting the European Union (EU) 2030 targets of pesticide-use reduction and the United Nations (UN) Sustainable Development Goals.

In this article, references to invertebrates include arthropods, such as insects and arachnids (including biocontrol agents such as mites and nematodes), but exclude crustaceans.

Table II provides a glossary of commonly used logistics terms for ease of understanding.

Time sensitivity and transport constraints

For the shipper, the most important goal is on-time service. Invertebrates are shipped at crucial species-specific life stages. Reactivity at collection is highly valuable, as is priority handling all along the transit chain.

A major barrier to on-time delivery is that rules for transporting live invertebrates are often unpredictable and inconsistent between countries. Larger IBCA suppliers and governmental and international organisations are generally familiar with the variations and can comply with shipping procedures. Yet even these larger organisations are not immune to sudden policy changes. For smaller suppliers and start-ups wanting to build reliable business based on shipping of invertebrates, the variation in requirements presents a significant stumbling block. Uncertainty and unreliability

Table II

Transport and logistics glossary

Terms and definitions extracted from the International Civil Aviation Organization–World Customs Organization Joint Publication on Air Cargo Security and Facilitation: Moving Air Cargo Globally (2016)

Aircraft operators	Aircraft operators, also known as airlines and air carriers, provide air transportation for goods. A transport contract (air waybill) binds an aircraft operator with the relevant contracted parties for the safe and secure transport of cargo and mail from one location (e.g. the airport of departure) to another (e.g. the airport of arrival). The air cargo may be transported on passenger aircraft or all-cargo aircraft. In some instances, particularly for short distances, aircraft operators may also transport air cargo by road. The transport contract nonetheless remains an air waybill, and the road segment is considered as a flight, with a designated flight number. This type of operation is known as a 'road feeder service'.
Air waybill	The air waybill is a document prepared by or on behalf of a shipper that evidences the contract between the shipper and aircraft operator(s) for the carriage of goods over routes of the operator(s). Air waybills have several purposes, but their two main functions are as a contract of carriage (behind every original air waybill are the conditions of contract for carriage) and as evidence of the receipt of goods.
Consignment	A consignment is one or more items accepted by the carrier from one shipper at one time and at one address, receipted in one lot and moving on one air waybill or shipment record to one consignee at one destination address.
Express carriers/ Integrators	Express carriers combine the work of a broker, haulier, freight forwarder, ground handler and aircraft operator into one single company or group, which is why they are also sometimes referred to as 'integrators'. Express delivery has become a specific business model in the cargo industry. Express carriers manage end-to-end multimodal supply chains. They operate sophisticated track-and-trace information technology systems, which allow them to monitor the progress of an individual shipment through their chain, from pick up to delivery. Express carriers typically transport high-value-added, time-sensitive cargo, with a time-definite delivery.
Freight broker	A freight broker is an independent agent who facilitates the movement of goods from buyer to seller, for instance by arranging air transport or meeting customs requirements, such as goods declarations. Brokering functions are often integrated with forwarding, consolidation and even warehousing functions within a single entity.
Freight forwarders	Freight forwarders are part of the transport logistics process within the supply chain. Their main task is to arrange for air shipments to be managed in such a way that they are ready for transportation by aircraft operators. Such arrangements might include the consolidation of cargo. A freight forwarder and logistics service provider may offer a service relating to the preparation, storage, carriage and final delivery of goods, including the applicable documentary and facilitation formalities. A forwarder rarely acts as carrier of the goods. A forwarder's prime responsibility is to safeguard the interests of its customer, the consignor or shipper.
Ground handlers	Ground handlers are subcontracted and act on behalf of freight forwarders or aircraft operators when the freight forwarder or aircraft operator does not have the necessary facilities. Ground handling services can include the provision of warehouses to accept, handle, prepare and tag cargo and mail, as well as loading, unloading, transit and storage of cargo and mail. Ground handlers are responsible for dealing with operational aspects, based on the instructions of freight forwarders and aircraft operators.
Known consignors	An entity may act as a known consignor when it originates cargo or mail for its own account and when its procedures meet common security rules and standards sufficient to allow the carriage of cargo or mail on commercial aircraft. Once a known consignor has accounted for the security status of cargo, the consignment may be delivered to a regulated agent, other approved entity or the aircraft operator, who then takes over the responsibility for keeping it secure until it is loaded onto an aircraft.
Receiver/ Recipient/ Consignee	The consignee is the party designated on the invoice or packing list as the recipient of the goods at the end of the transport movement.
Regulated agents	A regulated agent is a freight forwarder or any other entity (e.g. ground handler) that conducts business with an aircraft operator and provides security controls that the appropriate authority accepts or requires for cargo or mail. Once approved as a regulated agent, an entity may conduct security controls for cargo, including the screening of goods. An aircraft operator may receive cargo secured by a regulated agent, which accounts for the security status of consignments.
Shipper/ Sender/ Consignor	A consignor is the entity or individual who initiates the movement or transport of the goods. In other words, it is the sender. (A 'known consignor' [defined above] is an entity recognized in some states as meeting specified security requirements). The term 'shipper' is often used to describe the entity or individual who initiates the trade in goods. Consignor and shipper are separate roles but can be the same entity or individual.

are detrimental for R&D, as unique samples can be damaged when delayed.

A weak point in ensuring fast delivery is port-of-entry inspections, where documentation requirements and customs procedures may differ among countries, and even between different ports of entry in a single country. These procedures and failure to supply the right documentation can be sources of significant delay. In 2021, R&D institutions reported cases of insects shipped as pupae (the stage before adult emergence), in which delayed handling at transit airports, agencies or ports of entry snowballed and resulted in delivery outside the survival window. Figure 1 shows the impact of the extended transport duration on fruit flies; malformation at emergence and death were due to suffocation from insufficient air supply in the packaging. Such examples demonstrate the need to label live invertebrate packaging in ways that highlight priority handling requirements.

Additionally, airfreight is sometimes unavailable due to unforeseen situations, such as the flight restrictions during the COVID-19 pandemic, or when low-cost aircraft operators are prohibited from taking live cargo. This shortage highlights the need for alternatives: for instance, express carriers, also referred to as integrators, such as DHL, FedEx and UPS. These stakeholders have end-to-end multimodal supply chain capacities and existing logistics routes that could be exploited for more efficient transport of live invertebrates (Table III).

The case of sterile insect technique programmes

Access to transport has direct effects on biocontrol achieved through sterile insect technique (SIT) programmes. These programmes, implemented globally, require massive and frequent releases of sterilised live insects. Their examples provide clear case studies of how important efficient transportation is for the programmes' success.

Regular weekly transcontinental shipments of invertebrates have been efficient and successful for several large-scale operational SIT programmes [7]. A prime example, summarised by Wohlfarter [8], involved Entomon Technologies (Pty) Ltd, a semi-commercial codling moth (*Cydia pomonella*) SIT programme in the Western Cape Province, South Africa. During the 2010/11 release season, Entomon experienced rearing difficulties. As the release season was well underway, the supply interruption threatened the continued biocontrol programme for local apple and pear growers. Through the effective structures of the National Plant Protection Organisation of South Africa, a division of the National Department of Agriculture, Land Reform and Rural Development, an emergency import permit was granted within a week, and sterile moths were delivered from the Okanagan-Kootenay Sterile Insect Release (OKSIR) programme in Osoyoos, British Columbia, Canada. The permit additionally indicated a local priority shipping status similar to that of medical supplies. This status allowed speedy

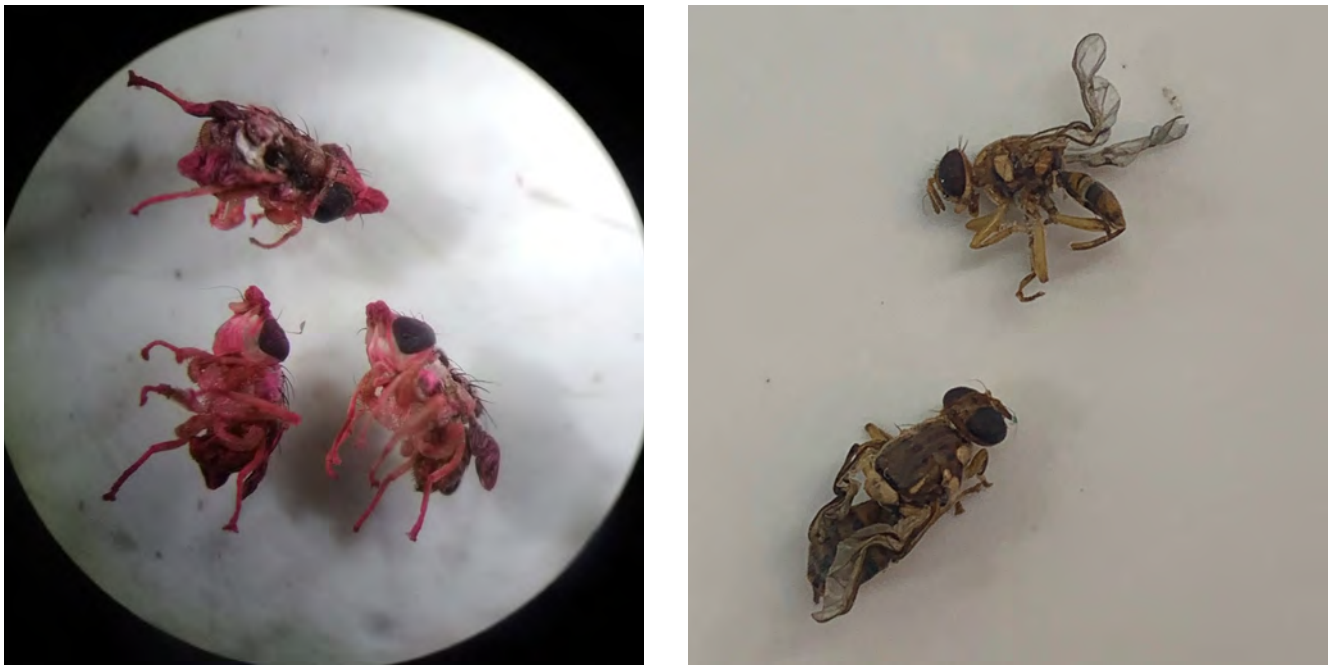

Figure 1

Malformation and mortality of sterile fruit flies shipped as pupae, due to the shippers' disregard for timely delivery

Left: *Ceratitis capitata* transported for >48h from Spain to Corsica (courtesy of Benoit Cailleret, Association de recherche et d'expérimentation sur fruits et légumes en Corse). Right: *Bactrocera dorsalis* transported for >5 days from Austria to Reunion Island (courtesy of Emma Dieudonné, Cirad)

Table III

How the main issues are currently dealt with and propositions for improvement

Issues	Current systems in place	Calls for improvements
Refusal of carriage by several express carriers	– The International Air Transport Association (IATA) Live Animals Regulations (LAR) describe some acceptable conditions for aircraft transportation. – Express carriers' policies to accept live insect cargo are country dependant. – Some national authorities have clarified that live insect transport should not fall under live animal transport requirements. [(a)]	– Increased awareness of sector importance and its positive contributions to protecting health and biodiversity. – Harmonisation of carriage conditions requirements from shippers for live insects. – Rapid access risk checklist to reduce uncertainty regarding packaging and handling; the LAR checklist [(b)] could be adapted to serve this purpose. – Live insect label with a clear destruction guideline.
Inefficient routes	– Freight forwarders and brokers optimise available routes by mixing transport modes, often resulting in high costs, several intermediaries and long transit times. – Dedicated road transport, often resulting in high costs and lengthy transport duration.	– Express carriers that manage end-to-end multimodal supply chains, thus securing tracking and time-sensitive deliveries. – Acceptance of live cargo by low-cost aircraft operators to benefit from more direct routes. – Greater integration and cooperation along the logistics chain to facilitate transit operations and ensure priority treatment.
High handling charges	– Insurance on liability is often a limiting element for express carriers. – Complex logistic routes involving several operators increase the cost.	– Clarified workflow and responsibility statements for each party. – Access express carriers that manage end-to-end multimodal supply chains, thus limiting intermediaries.
Unnecessary inspections	– Known consignor status facilitates the security clearance process on exporting cargo. – Documentation attached with packages explains contents and helps customs to handle the packages quickly. – Mutually recognised electronic certification systems facilitate speedy processing.	– Increased awareness of live insect packages requirements and risk mitigation by handling and customs agents. – Prior notification to customs inspectors. – Consistent documentation requirements for all invertebrates. – Live insect label with a clear checklist for accompanying documentation and authorisation.

(a) Example: Technique Instruction from the French Ministry of Agriculture and Food DGAL/SDSPA/2017-742 [5]
(b) LAR 47th Edition IATA Live Animals Acceptance Checklist [6]

customs processing and facilitated rapid road transport from the port of entry to the Entomon distribution facilities.

To date, very few SIT programmes operate in Europe, in part because emerging and promising projects have been impaired or stopped due to their reliance on commercial flights, which were unavailable during the COVID-19 pandemic. For example, the longstanding Mediterranean fruit fly (*Ceratitis capitata*) SIT protection programme, which imported sterile medflies to the Neretva Valley in Croatia, had to revert to conventional control (Luka Popović, Head of the Department for Research and Application of SIT, Croatian Agency for Agriculture and Food [Rukovoditelj Odjela, Odjel Za Israzivanje I Primjenu SIT Tehnike], personal communication). Dependence on logistics availability is a major impediment for customers. An experimental codling moth SIT project in southern France is making slow progress, as shipping from OKSIR was blocked by the lack of flights during the COVID-19 period (Box 1). Even prior to the pandemic, trial imports had low success rates due to handling delays at the port of entry. These attempts had to involve multiple freight forwarders, each only able to provide the appropriate logistics within either Canada or France. No single provider could be found to handle complete door-to-door service. Using multiple providers added intermediary costs, reduced efficiency of tracking and reactivity, complicated documentation and voided liability. In response, resources were diverted to develop local production capacity to circumvent import constraints. Such logistic limitations hamper rapid proof of concept and early adoption of biocontrol, at least in the short to mid-term.

The same challenges are also experienced by large commercial producers and international organisations on shipping routes between EU member states and from mainland Europe to outlying territories such as the French islands of Corsica in the Mediterranean Sea and Reunion in the Indian Ocean. Several European SIT pilot projects have been hampered or halted by late delivery, often because of non-priority handling that extended transit times. Shipping attempts using packages designated only as unspecified 'cargo' had random chances of timely delivery or suffered from poor handling. In some instances, brokers or freight forwarders proposed solutions that would allow timely delivery, but the costs were prohibitive for long-term operations.

Potential efficient and flexible alternatives to conventional air carrier operators and brokers include express carriers, or integrators, which use multiple modes of transport and can transport high-value-added or time-sensitive cargo,

Box 1

Shipping challenges that prohibit SIT initiatives from collaborating with the successful OKSIR model on production and release as an interim phase

The Okanagan-Kootenay Sterile Insect Release (OKSIR) Program in Canada employs sterile insects to control codling moths in pome fruit production. As a result, over the past 25 years, pesticide use has been reduced by 96% [9].

Sterile insects have been successfully shipped to South Africa and New Zealand for pilot SIT projects. But shipping costs prohibit expansion to regular commercial shipping, as these consignments are restricted to air cargo only and exclude couriers or other integrated options that cover the entire route with expedited delivery for perishable products. For emerging interests in European countries in particular, shipping costs have been prohibitive.

Policy has also hampered collaboration between OKSIR and a commercial partner in the United States of America to deliver SIT to growers. Although distribution on either side of the international border is handled by respective couriers, the actual cross-border shipping has only been practical when conducted by permitted hand-carriage.

Lastly, misalignment of transport- and certification-related policies between exporting and importing countries can cause a serious impasse. For import into New Zealand, codling moth consignments needed to be accompanied by a veterinary certificate. Yet, the mandated NPPO, the Canadian Food Inspection Agency, was unable to issue such documentation, as the procedure for lepidoptera was not covered by Canadian federal law.

and low-cost airlines, which traditionally have not carried live cargo but are increasingly opening new routes. Such alternatives are currently blocked in some cases, however, as some international express carriers maintain inconsistent rules that differ between countries, allowing live invertebrate transport on some routes, but prohibiting it on others.

Carrier concerns

Express carriers raise five primary concerns regarding the shipment of IBCAs. The first pertains to shipper insurance requirements related to timely delivery and the survival and quality of the specimens upon arrival. Although no shipper or receiver expects 100% fault-free services, they do expect that the fastest routing and processing is provided to ensure timely delivery, as with other perishable products. Shipping routes that involve transit through airports may face delays that are beyond the responsibility of the carrier and lead to either increased financial charges or viability loss of the shipment. Increased awareness by all stakeholders is crucial to ensure a streamlined service (Table III).

Secondly, low frequency shipments are not attractive for most carriers. But, taken collectively, the diversity of users (including research centres and industrial users) that are currently underserved should make this sector commercially attractive (Table I; Enkerlin and Pereira [7]; Oliva *et al*. [10], this issue).

Thirdly, express carriers often raise concerns about packaging. Some prohibit invertebrate transportation because they think the invertebrates might escape and cross-contaminate their vehicles, containers, and other products in the warehouses, or escape into the wild. These concerns seem to originate from misconceptions regarding the packaging prepared by the shipper, or confusion over the procedures for destroying and disposing undeliverable items. Prevention of escape is generally adequately handled by the shipper, to whom the security and quality of the consignment is of primary importance.

Fourthly, some express carriers are reluctant to guarantee transport when they cannot control ambient temperature conditions, especially during summer. However, it essentially remains the shipper's responsibility to prepare the parcel in tailored packaging that maintains conditions of low light, low temperature and suitable O_2:CO_2 balance, which can help preserve the climatic conditions required should the shipment be delayed. Similarly, this is achieved for shipping other fresh or sensitive commodities, such as food and vaccines.

'Known consignor' agreements (defined in Table II) establish the shipper as trusted and as an experienced safe exporter, which should reassure carriers. These agreements are recognised by the International Air Transport Association (IATA), and shippers receive certification by their respective national aviation security authority. The known consignor scheme streamlines security clearance of the packages before uplift onto an aircraft, but will not circumvent inspection during international transit or at the final port of entry. It still does not guarantee rapid processing for delivery. Known consignor packages are handled by certified regulated agents, which can be express carriers.

Finally, some express carriers refuse shipment because of unclear regulation concerning their authorisation to ship live invertebrates. Express carriers are strictly governed by internationally accepted guidelines on the transport of hazardous or dangerous goods. Invertebrate biological control agents are not classified under these categories. Guidelines for transporting non-hazardous biological products are not covered by either the European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways, the International Maritime Dangerous Goods Code or the Regulation concerning the International Carriage of Dangerous Goods by Rail.

Regarding air transport only, carriers can find guidance within the IATA Live Animals Regulations (LAR) [11]. This regulation provides a brief explanatory note on what IBCAs are and how they can be handled safely, including shipping and packaging processes that enable adequate confinement.

Commodities with similar requirements

Express carriers already have efficient logistic solutions in place for several kinds of fragile goods, such as vaccines and other perishables. Table IV draws a comparison between the requirements for vaccine shipment and those for invertebrates. This article shows that the capabilities and capacities already exist for vaccines and, therefore, argues that there is no reason why they cannot be used for invertebrates.

Consultation with these stakeholders could provide solutions to the issues encountered by live invertebrate shippers, as described in Table III.

Benefits to carriers

Large commercial IBCA manufacturers and distributors have traditionally relied on in-house logistics operations for short- to medium-haul shipments. With globalisation and an increasing demand for smaller volumes sent to an expanding number of locations, existing transport solutions have limited capacity or efficiency. Additionally, smaller start-up businesses with limited shipping frequency contribute to a growing need for transport flexibility. A 'business as usual' approach from carriers risks missing this growing business opportunity.

Table IV

A comparison of existing requirements between shipping vaccines and shipping invertebrate biological control agents (IBCAs)*, and how they could serve to improve further invertebrate shipping

Key steps for an optimal shipment	Vaccine shipments	IBCA shipments	How to improve invertebrate shipping
Prevent environmental and health risks: Safe packaging	Shipper is responsible for the vaccine to be packaged carefully, to ensure no breakage or leaks during shipment. Strict protocols apply.	Shipper is responsible for the IBCAs to be packaged securely to ensure their condition during shipment is maintained and that the package is secured against escape. Strict protocols are used in packaging invertebrates for shipment, as defined in the IATA LAR.	Extend the LAR protocol to other types of invertebrates. LAR protocols used in packaging invertebrates for successful shipment to address handling conditions.
Ensure quality and survival: Suitable transport conditions	Shipper is responsible for the vaccine to be packaged in such a manner that temperature extremes encountered during shipping will not degrade the vaccine.	Shipper is responsible for packaging the IBCAs in such a manner that temperature extremes encountered during shipping will not affect their viability.	
Ensure flawless logistics: Proper documentation	Proper documentation for approval to import the vaccine must be in place for customs inspection. Freight brokers or shippers are responsible to provide this to the carrier. Strict guidelines exist to determine and assign responsibilities at every step of the process to facilitate clearance through customs and transportation to the destination.	Proper documentation for approval to import the invertebrates (in the form of sanitary or phytosanitary certificates) must be in place for customs inspection. Freight brokers or the shippers are responsible to provide this to the carrier.	Through the International Plant Protection Convention and NPPOs and their respective competent authorities, regulations and protocols are in place to accommodate the approval and authorisation of imported IBCAs. Similar procedures can be followed for other invertebrates. Known consignor status or prior notification may facilitate quick passage at the exporting port. Clear guidelines are needed to facilitate subsequent steps of the process.
Ensure market access: Solutions for customers	Express carriers will accept vaccine for international shipment and delivery. The WHO Guidelines [12] mainly cover airfreighting.	Express carriers often prohibit shipment of invertebrates. Aircraft operators usually accept live IBCAs shipments and follow the LAR procedure.	Based on the protocols and regulatory approval processes in place, shipping invertebrates by express carriers or aircraft operators is no different from shipping vaccines and should be easily accepted. More solutions from express carriers are needed to avoid complex routes involving several operators. Handling of live cargo by low-cost airlines could also provide cheaper and faster route for brokers.

* The World Health Organization (WHO) [12] prescribes labelling, packaging and temperature-monitoring procedures for vaccines. Responsibilities of shippers and receivers are outlined. Various labels on delivery time and temperature range requirements are listed. Route and advance notification advice is also given. Packaging and monitoring device specifications are provided for many common vaccines. A similar document could be produced to provide model guidance for shipping insects

The ability of an express carrier to differentiate its service is becoming increasingly important to maintain competitiveness and customer loyalty. Much focus is placed on CO_2 emissions and biodiversity protection. Another opportunity is in the transport of IBCAs, which are used for agriculture, animal and human health protection, and conservation. Their shipment has a direct positive effect on the environment, as it enables the replacement of conventional pesticides in pest management programmes. The logistics sector should be surveyed to learn if there is interest in supporting such green initiatives.

Current limitations of live invertebrate import documentation

Various guidelines concern general and sector-specific procedures supporting safe and timely trade in live invertebrates. Shipping agents should have confidence that the shipper and receiver have adequately dealt with these requirements. These guidelines are, however, rarely fully understood by shippers and transport stakeholders.

International Standard for Phytosanitary Measures No. 3 [3] aims to ensure safe global movement of biocontrol agents and other beneficials, including their export, shipment, import and release. National plant protection organisations are advised to adopt these guidelines and accordingly communicate with and train local stakeholders. The same ISPM lists the type of documentation that might be required for trade in biocontrol and beneficial insects, but does not provide model templates. Phytosanitary certificates cannot be used in certain cases related to invertebrates: ISPM No. 12 [13] states that such certificates 'should not include statements that requirements have been met and should not include references to animal or human health matters', thereby limiting their application to some of the issues identified for insect trade. Other documentation can be linked or noted on a phytosanitary certificate, however. The scope of ISPM No. 3 is considered further by Quinlan *et al*. [14], in this issue.

The EPPO guidelines PM 6/1 [15], PM 6/2 [16], PM 6/3 [4] and PM 6/04 [17] align with the international standard. They have been largely adopted by the NPPOs in the 52 member countries of the regional plant protection organisations in Europe, Eastern Europe, the Middle East, Central Asia and Northern Africa.

Additionally, the International Atomic Energy Agency (IAEA) has extensively published on shipping guidelines to ensure safe transport of insects for SIT [18]. Cross references to ISPM No. 3, or even its inclusion as an addendum into IATA LAR guidance, might clarify the relationship among the different sources.

Regarding aircraft transport, Section 10.4.5 of the IATA LAR briefly describes procedures for IBCAs, particularly pointing out that IBCAs are 'thoroughly researched' and that 'there is no evidence of them causing problems for humans, animals and the environment' [11]. These statements are intended to provide carriers with assurance and clarity regarding their safety. The IATA LAR furthermore provides a 'Container Requirement Section', requirements 61–63, 65 and 67–68, with detailed suggestions on the criteria containers should meet for the safe transport of IBCAs, including pollinators.

Sanitary certification

Some authorities of importing countries (including customs and NPPOs as well as Veterinary Services) require sanitary certification confirming that invertebrate shipments are free from disease vectors and listed mammalian, avian and aquatic pests, even when used only for plant protection. These certifications are frequently required for shipments of bumblebees (*Bombus* spp.) and other invertebrates, despite not falling under World Organisation for Animal Health (WOAH, founded as OIE) guidance. The requirement of such certificates originates from misinterpretation of the WOAH *Terrestrial Animal Health Code* [19] (see Torres *et al*. [20] in this issue, on which insects are covered by WOAH) and IATA LAR guidelines.

Shipments within the EU originating from EU member states are exempt from certification requirements [21]. But occasionally, some member state authorities still erroneously insist on veterinary certification for shipments within the EU.

Blanket prerequisites can undermine the true value of veterinary certificates, as they end up being issued simply to tick the requirement box, rather than to provide real value. Certification of IBCAs against diseases such as foot and mouth, vesicular stomatitis, etc., is irrelevant as these beneficial insects are neither hosts nor vectors of these pathogens. Similarly, bumblebee consignments often require certification as being free from diseases that they do not host.

Some Central American, African and Asian countries require phytosanitary certification for IBCAs [21], which indicates a misinterpretation of the purpose of the certificate.

Moreover, under R&D activities, shipped invertebrates originating from the wild are often sent for identification or to confirm the presence of pathogens. In such cases, the process described under ISPM No. 3 is not as easy to follow due to unknown identity, unknown disease status and lack of related local capacity at the site of issuance or inspection.

Globally, the Agreement on the Application of Sanitary and Phytosanitary Measures (SPS Agreement) of the World Trade Organization provides for relevant certification of imports and exports, but guards against the use of certification as an artificial trade barrier [22]. Nonetheless, for some shippers, difficulties to obtain the appropriate documentation are indeed becoming a barrier to trade in live insects.

Inconsistency in this regard can result in an importing country requiring a particular sanitary certification that an exporting country may not issue. As an example, the Canadian impasse, described in Box 1, required that the federal parliament of Canada revise its export regulations to provide New Zealand with the required documentation before any shipment could commence.

Certification of invertebrates for biocontrol should be fit-for-purpose, confirming that the organisms in question are beneficials, that is, predators or parasitoids of agricultural or domestic pests, or, in the case of SIT, rendered safe through sterilisation treatments. Release authorisation from a national authority confirms that these invertebrates are safe and of no danger to non-target organisms. In the case of SIT, risks related to escape are removed by the conferred reproductive sterility and, for some species, male-only selection.

Sanitary and phytosanitary policy enforcement can rest with a variety of competent authorities such as NPPOs, agricultural ministries, customs and trade and industry authorities. As a result, many countries use mutually recognised electronic certification systems to speed the issuance of standardised documentation. Effective communication with carriers by the competent authorities could resolve carriers' current lack of procedural understanding.

Recommendation for specific labelling

Even when all documentation for import is in place, stakeholders that handle the packages along the shipping route may face procedural uncertainties, especially during transport delays or other unforeseen situations. Since July 2012, all healthcare-related product shipments are furnished with a Time & Temperature Sensitivity label in order to maintain consignment integrity [23]. This label allows the shipper to indicate the suitable temperature range of the consignment, marks the package as time-sensitive and alerts the carrier not to allow extensive re-routing or holding. It does not, however, indicate a preferred delivery window (time from sending to receiving).

Various senders already use a combination of privately produced labels for temperature parameters and rush transport. No single all-encompassing label exists specific to insects and arachnids. The authors propose such a dedicated label (Figure 2), which includes the following sections:

- a declaration of Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) or non-CITES status
- day and time sent
- last viable arrival date and time
- temperature range
- guidelines for safe destruction of the package, should it be necessary.

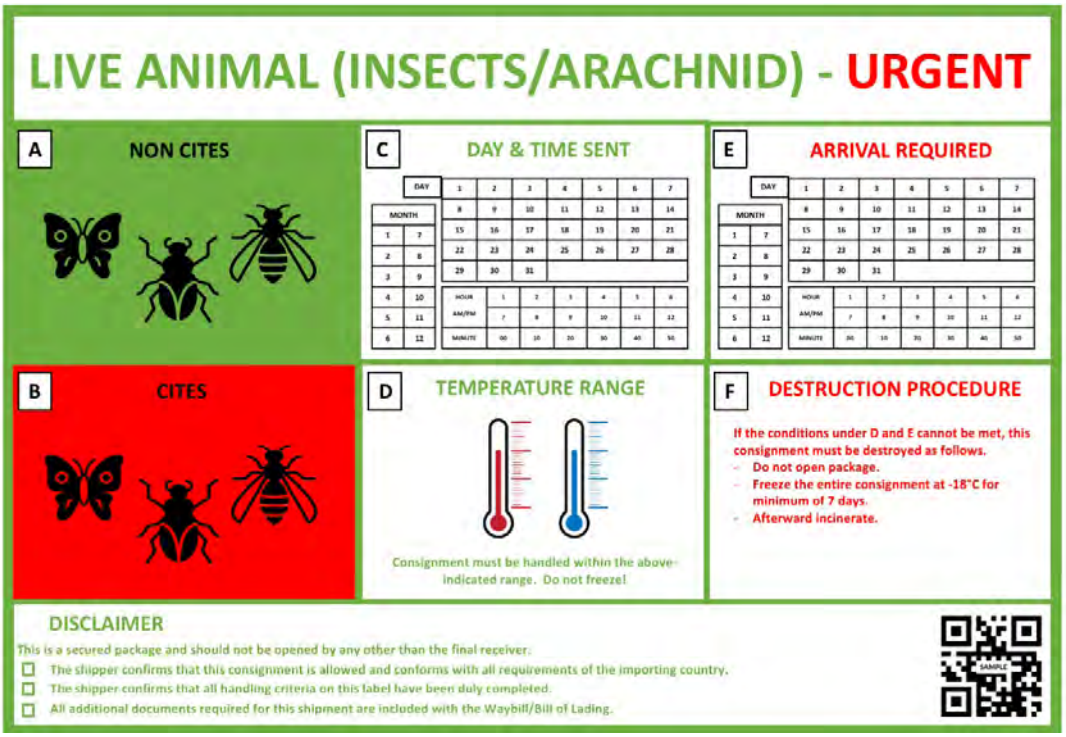

Figure 2

Proposal of an informative label for the shipment of insects or arachnids

Users would simply mark the relevant criteria and disclaimer, ensuring a truthful completion of the details (Figure 3). A QR code is also provided, which could be consulted by any user along the route to obtain more details on the various sections of the label.

Conclusions

The transport of live insects is impeded by four main problems: (1) unwillingness by some express carriers and integrators to transport, (2) lack of appropriate routes to allow timely delivery, (3) disproportionate costs compared to classic goods and (4) inconsistent inspection requirements.

Existing guidelines (ISPM No. 3, IATA LAR) address some of the challenges to invertebrate shipping, but increased awareness on the relevant requirements of live invertebrates is needed at all stages of the logistics and inspection chains. Suggested improvements entail global harmonisation of rules (while retaining useful distinctions between types of trade) and the creation of a universal checklist specific to this trade, combined with a live-invertebrate-specific label.

Regular and known consignor shipments are generally handled efficiently by freight forwarders and aircraft operators, but sometimes with limited availability of direct or compatible routes. Express carrier companies have experience in rapid transportation and careful handling of other types of sensitive or perishable products; they could offer significant shipping flexibility to the IBCA market thanks to their end-to-end multimodal logistic chain. If the buy-in of express delivery providers could be obtained, their integrated model could result in competitive transport pricing, addressing issues of routing and costs.

To date, several stakeholders are underserved. This underservice limits their R&D activities, commercial development and expansion of biocontrol approaches. The safe and efficient transport of IBCAs, pollinators and R&D material is in the interest of all stakeholders. For policy-makers and authorities, resolving this basic issue would boost the integration of biocontrol and other beneficial uses of insects, thus empowering scientific reactivity to pathogen transmission and risks posed by invertebrate invasions. Opening new routes and contributing to biocontrol deployment and biodiversity protection creates new opportunities for express carriers and integrators, and should be viewed as positively impactful, for example, promoting green label initiatives.

Well-communicated guidelines that confirm consignment safety should foster consistent inspection requirements and prevent unnecessary holdups at ports of entry that negatively affect the organisms' survival.

This is a call to the transport stakeholders to recognise the many procedures already put in place by competent authorities. Further discussions could be facilitated and

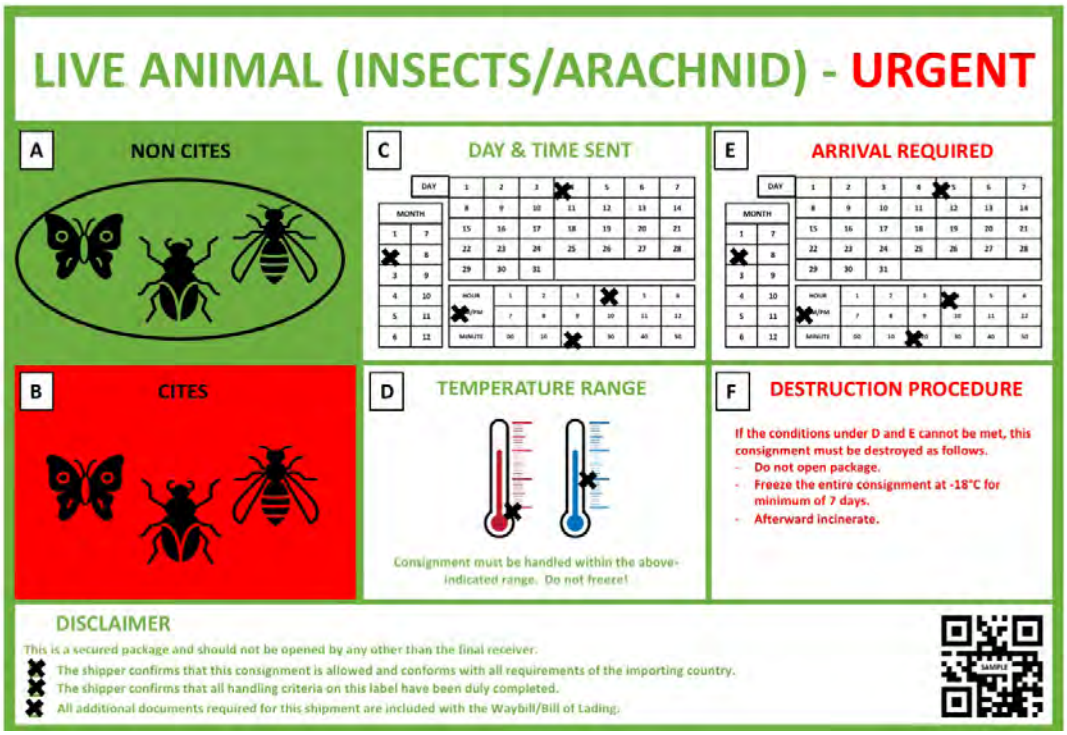

Figure 3

Filled-in example of the proposed label for the shipment of insects or arachnids

conveyed to stakeholders by leading international organisations such as WOAH, the IAEA, the International Plant Protection Convention and the International Biocontrol Manufacturers Association in consultation with competent national authorities. Their work could harmonise these sometimes-disparate guidelines and regulations into a more accessible framework for safe and efficient carriage of insects.

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Rationaliser les expéditions transfrontalières d'invertébrés vivants

M. Wohlfarter, C.F. Oliva & C. Nelson

Résumé

La logistique en appui des expéditions d'invertébrés vivants devrait être simple : elle doit veiller au respect des délais de livraison et s'assurer d'un conditionnement qui garantisse à la fois la survie des insectes et un confinement sans failles. Le conditionnement relève de la responsabilité de l'expéditeur, qui a tout intérêt à préserver la qualité du produit durant le transport et à veiller à ce qu'aucun spécimen ne s'échappe. Le respect des délais de livraison est déterminé par la capacité de l'opérateur à organiser un itinéraire de transport rapide et par l'efficacité mise en œuvre par les transporteurs dans l'exercice de leur mission.

Les auteurs examinent ce secteur sous-exploité. Alors que des capacités logistiques du même ordre sont déployées pour d'autres marchandises exigeant une livraison rapide (par exemple les vaccins et les produits alimentaires frais), les parties prenantes souhaitant organiser une expédition d'invertébrés vivants peinent souvent à trouver des transporteurs capables d'assurer cette prestation, voire acceptant de le faire. À partir de quelques exemples tirés de la lutte biologique, les auteurs font ressortir la complexité et les incohérences de la logistique mise en œuvre. Pour certains pays et acteurs, ce problème peut constituer un obstacle majeur à la croissance et à la durabilité du secteur de la lutte biologique.

Les auteurs relèvent également les idées fausses (concernant le conditionnement, la responsabilité légale et les contraintes administratives) et le manque de clarté réglementaire (par exemple l'obligation de fournir un certificat vétérinaire générique alors qu'un tel document est rarement pertinent pour des invertébrés), facteurs probablement à l'origine du refus de certaines entreprises de livraison express de transporter des invertébrés vivants. Ces problèmes ont pour conséquence que les lots ne sont pas traités de manière prioritaire pendant les liaisons de transport ou les procédures de dédouanement, ce qui entraîne des retards importants.

Les auteurs proposent des améliorations qui pourraient rationaliser ces transports en introduisant des changements adaptés aux procédures d'expédition en vigueur. En outre, les auteurs exhortent les acteurs du transport et de l'inspection à utiliser les orientations existantes et d'autres ressources disponibles afin d'apporter un soutien plus efficace à ce secteur sous-développé.

Mots-clés

Courtiers – Expédition – Invertébré vivant – Logistique – Lutte biologique – Transporteurs express.

Agilización del envío transfronterizo de invertebrados vivos

M. Wohlfarter, C.F. Oliva & C. Nelson

Resumen

Los aspectos logísticos ligados al envío de invertebrados vivos deberían estar meridianamente claros: lo que se requiere es una entrega puntual y un embalaje que garantice la supervivencia y el confinamiento de los animales. El embalaje es responsabilidad del expedidor, que es el primer interesado en que la calidad del producto no se resienta durante el envío y en que los especímenes no escapen. La puntualidad de la entrega depende de la capacidad del agente de expedición para organizar un itinerario adecuado y de la eficacia del transportista a la hora de cumplir su cometido.

Los autores describen este sector, hoy algo desatendido. Mientras que para otros bienes que requieren una entrega rápida (como vacunas o sangre fresca) existen tales medios logísticos, los profesionales que dependen del envío de invertebrados vivos tienen a menudo dificultades para encontrar transportistas que estén en condiciones o tengan la voluntad de asumir tales servicios. Los autores presentan ejemplos de control biológico que evidencian la complejidad e incoherencia de la logística que actualmente rodea estas actividades. Para ciertos países y partes interesadas, la cuestión puede llegar a suponer un importante freno al crecimiento de un sector de lucha biológica que sea sostenible.

Los autores exponen asimismo una serie de ideas equivocadas (sobre el embalaje, la responsabilidad o los trámites burocráticos) y de reglas confusas (como la exigencia de certificados veterinarios genéricos que rara vez son aplicables a los invertebrados) que pueden hacer que las empresas de transporte urgente se nieguen a transportar invertebrados vivos. Debido a estos problemas, ocurre a menudo que los bultos no sean considerados prioritarios en las conexiones de transporte o los despachos de aduanas, cosa que provoca importantes retrasos en las entregas.

Los autores proponen mejoras que podrían agilizar el transporte y a la vez encajarían en los procesos de envío tal como discurren actualmente. Además, con este artículo desean alentar a cuantas partes intervienen en las actividades de transporte e inspección a que utilicen las guías y demás recursos existentes para prestar un apoyo más eficaz a este infradesarrollado sector.

Palabras clave

Control biológico – Envío – Intermediarios – Invertebrado vivo – Logística – Servicios de transporte urgente.

References

[1] International Biocontrol Manufacturers Association (IBMA) (2021). – IBMA 2021 biocontrol industry survey calls on members to make a strong can-do statement. IBMA, Brussels, Belgium, 3 pp. Available at: https://ibma-global.org/wp-content/uploads/2021/04/2021-04-07-Press-Release-IBMA-Biocontrol-Industry-Survey-2021.pdf (accessed on 19 February 2022).

[2] Lamichhane J.R., Bischoff-Schaefer M. […] & Villeneuve F. (2017). – Identifying obstacles and ranking common biological control research priorities for Europe to manage most economically important pests in arable, vegetable and perennial crops. *Pest Manag. Sci.*, **73** (1), 14–21. https://doi.org/10.1002/ps.4423

[3] Food and Agriculture Organization of the United Nations (FAO) (2017). – Guidelines for the export, shipment, import and release of biological control agents and other beneficial organisms. International Standards for Phytosanitary Measures (ISPMs), ISPM No. 3. FAO, Rome, Italy, 16 pp. Available at: https://www.fao.org/3/j5365e/J5365E.pdf (accessed on 1 February 2022).

[4] European and Mediterranean Plant Protection Organization (EPPO) (2021). – PM 6/3 (5) biological control agents safely used in the EPPO region. *EPPO Bull.*, **51** (3), 452–454. https://doi.org/10.1111/epp.12801

[5] Ministry of Agriculture and Food (France) (2017). – Instruction technique DGAL/SDSPA/2017-742. Ministry of Agriculture and Food, Paris, France, 13 pp. Available at: https://oaba.fr/PDF/instruction%20DGAL2017_rappel_reglementation.pdf (accessed on 1 February 2022).

[6] International Air Transport Association (IATA) (2021). – IATA Live Animal Acceptance Checklist. *In* Live Animal Regulations. 47th Ed. IATA, Montreal, Canada, 230–231. Available at: https://www.iata.org/contentassets/d7c512eb9a704ba2a8056e3186a31921/en-lar-47-acceptance-checklist.pdf (accessed on 1 February 2022).

[7] Enkerlin W.R. & Pereira R. (2022). – The sterile insect technique: an international framework to facilitate transboundary shipments of sterile insects. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech*, **41** (1), 66–74. https://doi.org/10.20506/rst.41.1.3303

[8] Wohlfarter M. (2012). – Entomon trials and tribulations. Unpublished report. Entomon Technologies (Pty) Ltd, Stellenbosch, South Africa, 17 pp.

[9] Nelson C., Esch E., Kimmie S., Tesche M., Philip H. & Arthur S. (2021). – Putting the sterile insect technique into the modern integrated pest management toolbox to control the codling moth in Canada. *In* Area-wide integrated pest management: development and field application (J. Hendrichs, R. Pereira & M.J.B. Vreysen, eds), 1st Ed. CRC Press, Boca Raton, United States of America, 111–127. https://doi.org/10.1201/9781003169239-7

[10] Oliva C.F., Chand R., Prudhomme J., Messori S., Torres G., Mumford J.D., Deme I. & Quinlan M.M. (2022). – International live insect trade: a survey of stakeholders. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech*, **41** (1), 29–65. https://doi.org/10.20506/rst.41.1.3302

[11] International Air Transport Association (IATA) (2022). – Live Animal Regulations. 48th Ed. IATA, Montreal, Canada, 548 pp.

[12] World Health Organization (WHO) (2020). – Guidelines for the international packaging and shipping of vaccines. 6th Ed. WHO, Geneva, Switzerland, 169 pp. Available at: https://apps.who.int/iris/handle/10665/338012 (accessed on 1 February 2022).

[13] Food and Agriculture Organization of the United Nations (FAO) (2017). – Phytosanitary certificates. International Standards for Phytosanitary Measures (ISPMs), ISPM No. 12. FAO, Rome, Italy, 30 pp. Available at: https://www.fao.org/3/k5129e/k5129e.pdf (accessed on 1 February 2022).

[14] Quinlan M.M., Mumford J.D., Messori S., Enkerlin W.R., Shimura J., Smith L., Dass B., Oliva C.F., Nelson C., Chand R. & Torres G. (2022). – Issues and gaps in international guidance and national regulatory systems affecting international live insect trade. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech*, **41** (1), 198–210. https://doi.org/10.20506/rst.41.1.3317

[15] European and Mediterranean Plant Protection Organization (EPPO) (1999). – PM 6/1 (1) First import of exotic biological control agents for research under contained conditions. *EPPO Bull.*, **29** (3), 271–272. https://doi.org/10.1111/j.1365-2338.1999.tb00831.x

[16] European and Mediterranean Plant Protection Organization (EPPO) (2014). – PM 6/2 (3) Import and release of non-indigenous biological control agents. *EPPO Bull.*, **44** (3), 320–329. https://doi.org/10.1111/epp.12153

[17] European and Mediterranean Plant Protection Organization (EPPO) (2018). – PM 6/04 (1) Decision-support scheme for import and release of biological control agents of plant pests. *EPPO Bull.*, **48** (3), 352–367. https://doi.org/10.1111/epp.12495

[18] Food and Agriculture Organization of the United Nations (FAO), International Atomic Energy Agency (IAEA) & United States Department of Agriculture (USDA) (2019). – Product quality control for sterile mass-reared and released tephritid fruit flies. Version 7.0. IAEA, Vienna, Austria, 148 pp. Available at: https://www.iaea.org/sites/default/files/qcv7.pdf (accessed on 21 February 2022).

[19] World Organisation for Animal Health (WOAH) (2021). – Terrestrial Animal Health Code. WOAH, Paris, France. Available at: https://www.oie.int/en/what-we-do/standards/codes-and-manuals/terrestrial-code-online-access (accessed on 26 October 2021).

[20] Torres G., Diaz F., Okamura Y., Messori S. & Hutchison J. (2022). – The World Organisation for Animal Health – current and potential roles in safe international trade of bees and other insects. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech*, **41** (1), 211–218. https://doi.org/10.20506/rst.41.1.3318

[21] Barratt B.I.P., Colmenarez Y.C., Day M.D., Ivey P., Klapwijk J.N., Loomans A.J.M., Mason P.G., Palmer W.A., Sankaran K.V. & Zhang F. (2021). – Regulatory challenges for biological control. *In* Biological control: global impacts, challenges and future directions of pest management (P.G. Mason, ed.), 1st Ed. CSIRO Publishing, Melbourne, Australia, 166–196.

[22] Bozzini E. (2017). – Pesticide policy and politics in the European Union: regulatory assessment, implementation and enforcement. Palgrave Macmillan, Cham, Switzerland, 120 pp. https://doi.org/10.1007/978-3-319-52736-9

[23] International Air Transport Association (IATA) (2013). – Time & Temperature Sensitive label FAQ. IATA Cargo, Montreal, Canada, 5 pp. Available at: https://www.iata.org/contentassets/6d7404d9ccca4e4e9c4ce146e4a2acb1/time-temperature-faq.pdf (accessed on 13 November 2021).

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